Models	Accuracy	Precision	Recall	F1	Cohen Kappa
Zero Shot Setting					
Zero Shot	0.68	0.76	0.64	0.65	-
+ improved prompt using Anthropic prompt generator	0.68	0.70	0.64	0.64	-
Prompt Optimizers					
DSPy					
- with idea, most relevant papers, class	0.68	0.83	0.62	0.58	-
- with idea, most relevant papers, class, reasoning	0.66	0.82	0.58	0.52	
TextGRAD					
- with idea, most relevant papers, class	0.78	0.76	0.76	0.76	-
In-context Setting					
Open-Review Examples					
- with idea & review (i.e., reasoning)	0.59	0.55	0.51	0.43	-
Expert Labeled Examples					
- with idea, reasoning	0.75	0.76	0.77	0.75	-
- with idea, most relevant papers, class	0.78	0.77	0.76	0.77	-
- with idea, most relevant papers, class, reasoning	<u>0.81</u>	<u>0.84</u>	<u>0.78</u>	<u>0.79</u>	<u>0.59</u>
Other Novelty Checkers					
AI Scientist (Lu et al.)	0.47	0.55	0.53	0.44	0.05
AI Researcher (Si et al.)					
- GPT-4O	0.78	0.81	0.74	0.75	0.52
- CLAUDE-3-5-SONNET	0.56	0.63	0.61	0.56	0.19

Table 1: Experimental Results using gpt-4o on expert-annotated dataset.

re-ranking component, allowing us to assess the importance of facet-based re-ranking; (iii) **Embedding Filtering:** Omits the LLM re-ranker entirely, relying only on the embedding-based filtering. This setup allows us to assess the importance of the LLM re-ranking step; and (iv) **Snippet Retrieval** and **Keyword Retrieval**: Each of these setups returned the top-k documents from their respective retrieval method (without embedding filtering or any LLM re-ranking), leveraging the inherent ranking/scoring provided by Semantic Scholar. This setup allows to assess the importance of both re-ranking steps. This structured setup enabled us to isolate the contribution of each component (retrieval method vs. re-ranking strategy) and evaluate whether they collectively brought key papers for novelty assessment into the top 10. We use o3-mini for evaluating novelty (Step 3) and gpt-4o for re-ranking (Step 2).

Classification Analysis: Table 2 shows that the complete system, which employs facet-based re-ranking in RankGPT, significantly outperforms its ablated variants in accuracy. The results demonstrate that methods relying only on keyword or snippet-based retrieval have much lower accuracy, and even alternate re-ranking strategies with a single embedding-based reranker or both embedding and general relevance RankGPT are insufficient to consistently bring key papers into the most relevant paper set. These findings show that combining facet-based reranking with embedding is critical for identifying the most relevant papers.

Table 2: Accuracy of predicting "not novel".

Method	Accuracy
Complete System	89.66%
- Relevance RankGPT	13.79%
- Embedding Filtering	10.34%
- Snippet Retrieval	8.62%
- Keyword Retrieval	5.17%

Analysis of the Most Relevant Papers: Table 3 compares the top-10 most relevant papers retrieved under each ablation setting with those from the complete system. Approximately 30% of the papers differ when using either embedding-based or general relevance RankGPT. Additionally, notable rank shifts are observed between the facet-based and relevance-based LLM rerankers. In contrast, without the reranking steps, both snippet and keyword retrieval exhibit minimal overlap with the final system's top results, highlighting the importance of the reranker stage.

6.3 Qualitative Analysis

Table 4 in the Appendix shows examples from our training set, including an idea, its most relevant papers, and the corresponding expert reasoning. While assessing novelty, we add both the titles and abstracts of the most relevant papers for each idea.

Figures 2, 3, and 4 in the Appendix qualitatively compare novelty evaluations by AI Scientist, AI Researcher, and Idea Novelty Checker (ours) on

Table 3: Comparing rank and overlap in retrieved papers with each variant to the complete system. *Overlap* indicates how many papers overlap on average with the complete system top-10 papers. *Rank Shift* measures the average absolute difference in rank positions (only among overlapping papers).

Method	Overlap (↑)	Rank Shift (↓)
Relevance RankGPT	7.97	0.67
Embedding Filtering	7.93	0.84
Snippet Retrieval	2.88	1.85
Keyword Retrieval	1.17	1.39

two research ideas. Idea Novelty Checker provides concise justifications for its novelty decisions by referencing key similarities and differences with existing works. For example, in Example 1, it correctly identifies the idea as 'novel' by highlighting these aspects. In contrast, AI Researcher evaluates each paper individually, classfying an idea as 'not novel' if any paper is considered citable; but in our examples, none of the papers were flagged as citable despite sharing similar purposes, leading to a 'novel' classification. Due to space constraints, we show insights only from the first paper for each example. Figure 4 indicates that while AI Scientist's judgments generally align with the ground truth and offer actionable suggestions, it sometimes misinterprets the idea—as in Figure 3, where its focus shifts from the idea to the accompanying code.

6.4 Prompt Sensitivity

In our experiments with TextGrad, we investigated how specific prompt instructions influence an LLM's ability to classify the novelty of an idea. Figures in Appendices 5, 6, and 7 present the accuracy of various prompts optimized with TextGrad on our dataset (train=25, validation = 10, test = 32).

Prompts with both non-zero and zero validation accuracy included various instructions for evaluating the novelty of ideas, such as assessing the uniqueness of methods and their comparison to existing research. Through this prompt optimization process, we observed interesting ways in which LLMs may evaluate novelty, like considering historical context, frequency of similar studies, comparative analysis with existing works, examining arguments for both novel and non-novel perspectives. However, prompts without these specific instructions also influenced accuracy, suggesting the complexity of novelty evaluation with LLMs.

Notably, some prompts with similar instructions showed different performance on validation data. For example, both prompt 3 (accuracy = 0) and prompt 9 (accuracy = 0.6) include instructions to evaluate if the idea introduces unique methodologies, and how it compares to existing work. However, the difference in their performance suggests that subtle variations in wording and instruction framing can significantly impact the classification performance. It remains unclear why certain prompts perform better despite having similar instructions.

Our analysis highlights the LLM's sensitivity to prompt design when assessing novelty of an idea. Even minor variations in wording and structure can lead to substantial performance changes, emphasizing the need for careful prompt engineering and well-chosen in-context examples to guide the LLM for idea novelty evaluation.

7 Conclusion

In this work, we propose Idea Novelty Checker a retrieval-augmented pipeline for evaluating the novelty of scientific ideas and generating literature-grounded rationales. Our formative study highlighted two main challenges in evaluating novelty: (1) retrieving the most relevant papers from a vast corpus, and (2) establishing a fixed notion of novelty due to its inherent subjectivity. To address the latter, we incorporate expert-annotated examples in our novelty checker where we consider an idea to be novel within a given topic domain if it (1) differs from all retrieved papers in at least one core facet—namely, purpose (a new objective), mechanism (a distinct technical approach), or evaluation (a distinct validation method); (2) uniquely combines these facets; or (3) applies them to a new application domain.

Our experiments on an expert-annotated dataset demonstrate that Idea Novelty Checker outperforms two well-known recent baselines, and our ablation studies confirm the importance of each component in our system. Furthermore, qualitative comparisons and analyses of prompt sensitivity provide additional insights into novelty evaluation.

8 Limitations & Future Work

While Idea Novelty Checker is superior in many aspects, it also has some limitations. For instance, due to context size constraints (with fifteen in-context examples for both *novel* and *not novel* categories), our analysis is restricted to the top 10 retrieved papers, which may disproportionately influence the overall novelty assessment. Additionally, our definition of novelty relies on expert annotations, and the same annotators who provided the in-context examples also classified the test ideas. This could potentially give our approach an advantage in understanding our view of novelty. Moreover, many of the ideas used for testing were generated by the same system (Radensky et al.) that produced the in-context examples, although some ideas were sourced from OpenReview.

In future work, we aim to address these limitations by expanding the literature scope using tools such as OpenAI Deep Research and ScholarQA (Singh et al.) and further refining our novelty evaluation to view novelty as a continuum rather than binary classification.

References

Abdelrahman Abdallah, Bhawna Piryani, Jamshid Mozafari, Mohammed Ali, and Adam Jatowt. Rankify: A comprehensive python toolkit for retrieval, re-ranking, and retrieval-augmented generation. Publisher: arXiv Version Number: 3.

Faez Ahmed, Sharath Kumar Ramachandran, Mark Fuge, Samuel Hunter, and Scarlett Miller. Interpreting idea maps: Pairwise comparisons reveal what makes ideas novel. 141(2):021102.

Davide Baldelli, Junfeng Jiang, Akiko Aizawa, and Paolo Torroni. TWOLAR: A TWO-step LLM-augmented distillation method for passage reranking. 14608:470–485. Book Title: Advances in Information Retrieval ISBN: 9783031560262 9783031560279 Place: Cham Publisher: Springer Nature Switzerland.

Joeran Beel, Min-Yen Kan, and Moritz Baumgart. Evaluating sakana's AI scientist for autonomous research: Wishful thinking or an emerging reality towards 'artificial research intelligence' (ARI)? Publisher: arXiv Version Number: 2.

Nicolas Bougie and Narimasa Watanabe. Generative adversarial reviews: When LLMs become the critic. Publisher: arXiv Version Number: 1.

Joel Chan and Joseph Chee Chang. 31 SOLVENT : A mixed initiative system for finding analogies between research papers.

DaEun Choi, Sumin Hong, Jeongeon Park, John Joon Young Chung, and Juho Kim. CreativeConnect: Supporting reference recombination for graphic design ideation with generative AI. pages 1–25. Conference Name: CHI '24: CHI Conference on Human Factors in Computing Systems ISBN: 9798400703300 Place: Honolulu HI USA Publisher: ACM.

Arman Cohan, Sergey Feldman, Iz Beltagy, Doug Downey, and Daniel Weld. SPECTER: Document-level representation learning using citation-informed transformers. pages 2270–2282. Conference Name: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics Place: Online Publisher: Association for Computational Linguistics.

Steffen Eger, Andreas Rücklé, and Iryna Gurevych. Pitfalls in the evaluation of sentence embeddings. pages 55–60. Conference Name: Proceedings of the 4th Workshop on Representation Learning for NLP (RepL4NLP-2019) Place: Florence, Italy Publisher: Association for Computational Linguistics.

Jingtong Gao, Bo Chen, Xiangyu Zhao, Weiwen Liu, Xiangyang Li, Yichao Wang, Wanyu Wang, Huifeng Guo, and Ruiming Tang. LLM4rerank: LLM-based auto-reranking framework for recommendations. pages 228–239. Conference Name: WWW '25: The ACM Web Conference 2025 ISBN: 9798400712746 Place: Sydney NSW Australia Publisher: ACM.

Tarun Gupta and Danish Pruthi. All that glitters is not novel: Plagiarism in AI generated research. Publisher: arXiv Version Number: 2.

José Manuel Gómez-Pérez, Andrés García-Silva, Rosemarie Leone, Mirko Albani, Moritz Fontaine, Charles Poncet, Leopold Summerer, Alessandro Donati, Ilaria Roma, and Stefano Scaglioni. Artificial intelligence and natural language processing and understanding in space: A methodological framework and four ESA case studies. Publisher: arXiv Version Number: 2.

Ishika Joshi, Simra Shahid, Shreeya Manasvi Venneti, Manushree Vasu, Yantao Zheng, Yunyao Li, Balaji Krishnamurthy, and Gromit Yeuk-Yin Chan. CoPrompter: User-centric evaluation of LLM instruction alignment for improved prompt engineering. pages 341–365. Conference Name: IUI '25: 30th International Conference on Intelligent User Interfaces ISBN: 9798400713064 Place: Cagliari Italy Publisher: ACM.

Hyeonsu B Kang, David Chuan-En Lin, Nikolas Martelaro, Aniket Kittur, Yan-Ying Chen, and Matthew K. Hong. a. BioSpark: An end-to-end generative system for biological-analogical inspirations and ideation. pages 1–13. Conference Name: CHI '24: CHI Conference on Human Factors in Computing Systems ISBN: 9798400703317 Place: Honolulu HI USA Publisher: ACM.

Hyeonsu B. Kang, Xin Qian, Tom Hope, Dafna Shahaf, Joel Chan, and Aniket Kittur. b. Augmenting scientific creativity with an analogical search engine. 29(6):1–36.

Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T.